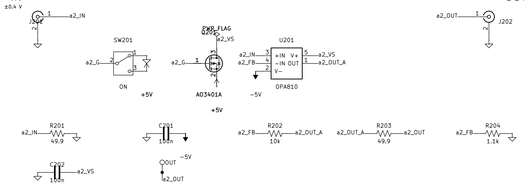
a2 x10, 14 MHz

cell (0,0)

tutorial The speed of sound at 40 kHz, Warming a crystal (+−5 V)

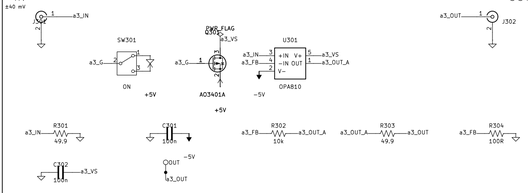


EXPECT: x10 (20 dB) into a high impedance, x5 into 50 ohm, −3 dB near 14 MHz; clips at about +−4 V
Non-inverting OPA810: gain 1 + 10 k / 1.1 k = 10.1, bandwidth 140 MHz / 10 = 14 MHz. 50 ohm at the input, 49.9 ohm
in series with the output (a 50 ohm load halves it, x5). Keep the input under 0.4 V peak: the output clips at 4 V.

a3 x100, 1.4MHz

cell (1,0)

tutorial The speed of sound at 40 kHz, Warming a crystal (+−5 V)

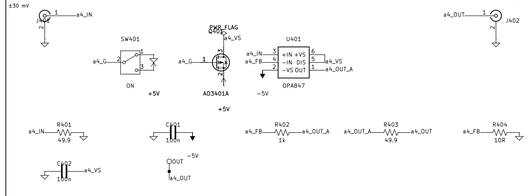


EXPECT: x100 (40 dB) into a high impedance, −3 dB near 1.4 MHz; keep the input under 40 mV peak
Non-inverting OPA810: gain 1 + 10 k / 100 = 101, bandwidth 140 MHz / 100 = 1.4 MHz. 50 ohm at the input, 49.9 ohm
in series with the output. For 40 dB out to 30 MHz use a4 (the OPA847 LNA); this tile is the FET-input, high-impedance
friendly x100 for everything slower. (The plan's two stages of x10 did not fit the cell.)

a4 LNA, 30 MHz

cell (2,0)

tutorial The speed of sound at 40 kHz, What time is it over there? (+−5 V)

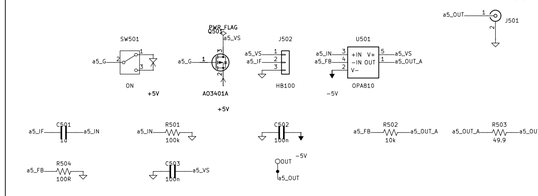


EXPECT: x100 (40 dB) from 50 ohm, flat 1–30 MHz, about 3 dB noise figure from 50 ohm; keep the input under 30 mV peak
Non-inverting OPA847 (3.9 GHz GBW at this gain, 0.85 nV/√Hz); 1 k / 10 ohm, so x101 and about 39 MHz. Its
input noise with the 50 ohm termination's own noise gives a noise figure near 3 dB. The op amp is stable only at
gains above 12 (it is decompensated); do not change Rg upward. SOT-23-6, DIS tied to +VS (enabled).

a5 Doppler, HB100

cell (3,0)

tutorial With a little more hardware (+−5 V)

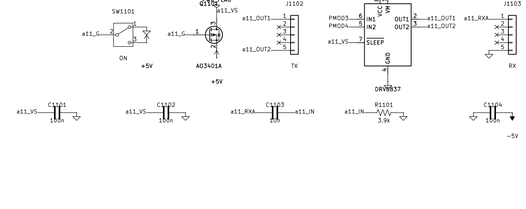


EXPECT: the Doppler beat of a target moving toward or away: 70 Hz per m/s (19.4 Hz per km/h), tens of mV after x101
HB100: a 10.525 GHz oscillator and mixer on one board; its IF is the beat between what it sends and what comes
back, 2v/lambda = 70 Hz per m/s. Wire its +5, IF and GND pins to the header; point it at something that moves.
The host's x10 input after this tile's x101 brings a hand at 1 m to the ADC's range.

a6 1-bit ADC

cell (0,1)

tutorial Trading speed for bits (+3.3 V)

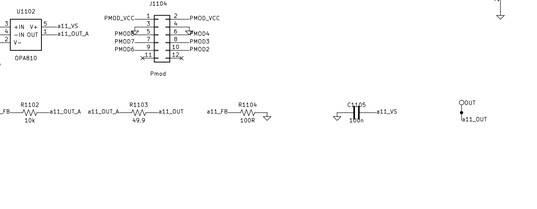


EXPECT: a one-bit stream on Pmod IO7 whose density follows IN: the FPGA closes the loop through Pmod IO8 (the handbook's gateware)
First-order delta-sigma: the comparator decides, the FPGA registers the bit and feeds it back through 10 k into the
10 k / 10 nF integrator (1.6 kHz; clock the loop at 1–10 MHz). IN takes 0 to 3.3 V, or +−1.5 V about the 1.65 V reference
if you AC-couple. Both Pmod lines are 3.3 V logic; the tile runs on the 3.3 V rail.

a7 PLL 4046

cell (1,1)

tutorial A faster DAC: PLLs (+5 V)

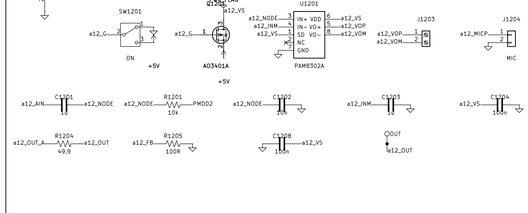


EXPECT: VCO OUT locks to a 50–200 kHz square or sine on IN; pin 9 is the demodulated FM, pin 13 the phase detector
74HC4046A: R1 10 k, C1 1 nF put the VCO near 100 kHz (R2 open: it can go down to 0). PC2 (the edge-triggered
tri-state detector) through 10 k / 100 n into VCOin: a lag filter, so expect ringing on lock. IN self-biases
through 100 n / 1 k (any level above 0.4 V peak). VCO OUT is 5 V CMOS through 470 ohm. Probe pins 9 and 13.

a8 heat

cell (2,1)

tutorial Warming a crystal. Control at the speed of the cable (+5 V)

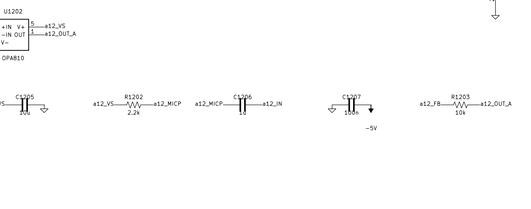


EXPECT: 0.83 W into the board when Pmod IO6 is high (the FET's drain), 10 mV/K + 500 mV on OUT
Two 15 ohm in series: 30 ohm, 170 mA, 0.83 W from the switched 5 V when Pmod IO6 is high (10 k keeps it off otherwise).
The TMP235 reads 750 mV at 25 C and climbs 10 mV per kelvin. A feedback-control experiment: hold the TMP235 at a
set point with the FPGA's PWM. (The plan's NTC divider did not fit the cell beside the two 2512s.)

a1 power, USB-C 5 V in: +5, −5, +3.3 out

cell (3,1)

tutorial A counter on the LEDs



EXPECT: +5 V from the USB-C (up to 1.5 A), −5 V at 40 mA, +3.3 V at 300 mA; the green LED says 5 V is in, the red one that −5 V is up
USB-C power only: 5.1 k on each CC asks any source for 5 V. A 2 A PTC, a 5 V TVS (clamps at 9.2 V) and 20 uF.
The LM2664 charge pump makes −5 V for the op-amp tiles (40 mA; four tiles at 10 mA); it is not switched per tile, an op
amp with its +5 V removed draws nothing. The TPS7A2033 makes 3.3 V for the comparator and the lines that meet the FPGA.
The socket along the top carries +5, ground, −5, ground, +3.3: the bench's rails for a breadboard experiment.

a9 mixer, MC1496

cells (0,2), (1,2)

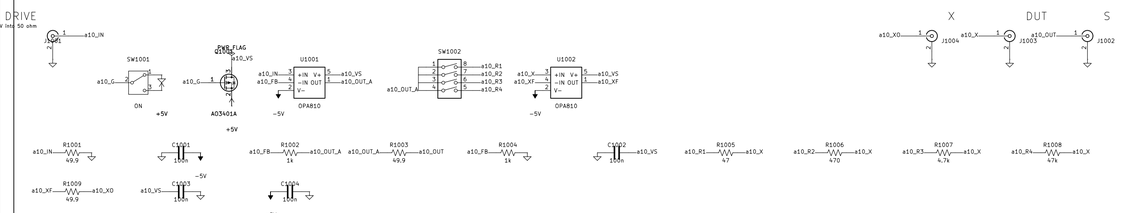
tutorial An AM radio, PSK and QPSK (+−5 V)

EXPECT: IF OUT = RF x LO (sum and difference frequencies, about 0.3 V peak per volt of RF at 0.3 V of LO); a balanced mixer, the carrier mostly cancels
Gilbert cell on +−5 V: signal pair at ground (51 ohm), carrier pair at 2.5 V (51 ohm to the 10 k / 10 k bias), 1 mA
per side (3.9 k), 1.5 k loads. Keep RF under 0.3 V and LO near 0.3 V peak (above that the carrier pair just switches,
which is fine for a product detector). IF OUT is single-ended from pin 6 through 100 n: a 1.5 k source, so read it
with a high-impedance input, not 50 ohm. No carrier-null trimmer: expect −40 dB of LO feedthrough.

a10 impedance, V and I

cells (2,2), (3,2)

tutorial RC and LC circuits, More ideas for one board (+−5 V)



EXPECT: S OUT and X OUT (each half the node's voltage into 50 ohm): Z = R X / (S − X); +−4 V and up to 40 mA for an I-V curve
DRIVE in ×2 is S. The DIP picks R between S and X (47 ohm for LEDs and diodes, 470 for most parts, 4.7 k and 47 k for
high impedances); the part goes on the OUT jack. Calibrate with the OUT open and shorted. An LED's knee against
its colour (±12) gives Planck's constant; a crystal (p8) shows its series and parallel resonances.

a11 ultrasound

cells (0,3), (1,3)

tutorial The speed of sound at 40 kHz (+−5 V)

EXPECT: a 40 kHz burst from Pmod IO4/IO3 leaves the TX transducer; its echo arrives x100 on RX OUT (hold a reflector over the board)
Drive IN1/IN2 (Pmod IO4, IO3) in antiphase at 40 kHz for 8–20 cycles; the bridge puts 10 V peak to peak across the TX
transducer. The RX transducer's 10 nF / 3.9 k high-pass (4 kHz) feeds an OPA810 at x100 (1.4 MHz: the transducer's
own resonance sets the band). Time of flight at 343 m/s: 5.8 us per mm there and back. Put the two transducers
in pins 1 and 5 (10.16 mm: their leads are 10 mm apart). The slide switch's 5 V powers the bridge and the op amp's +rail.

a12 audio

cells (2,3), (3,3)

tutorial ADC samples to Python, MSK and GMSK (+−5 V)

EXPECT: a speaker on SPK follows SPEAKER IN (2.5 W into 4 ohm) or the one-bit stream on Pmod IO2 (10 k / 10 n): MIC OUT is the electret x100
PAM8302A: 1 u couples SPEAKER IN to IN+ (IN− grounded through 1 u), the one-bit DAC on Pmod IO2 arrives through 10 k
into 10 n (1.6 kHz) at the same node; so both play. SD high: on with the tile (pin 2 is NC). The electret (pin 1 is the
+ side, 2.2 k from VS) through 1 u into an OPA810 at x100: a few hundred millivolts for speech.

Drawn by scripts/gen_sch.py from boards/amod_a.py: the expected responses from sim/passive.py
Every tile is one circuit with its own SMA jacks; the board's silkscreen draws each tile's wires as they
gain blocks, an HF LNA, a 10.5 GHz Doppler radar, a one-pin ADC, a 74HC4046 PLL, heat, a mixer, a
Board: Jason Gallicchio and Claude | open hardware, CERN-OHL-P-2.0 | github.com/gallicchio/amod

Sheet: /	
File: amod_a.kicad_sch	
Title: AMOD-A: active building blocks on USB-C power	
Size: A1	Date: 2026-10-25 — **Rev: A**
KiCad E.D.A. 10.0.7	Id: 1/1